7. Other regression techniques
Weighted Least Squares

In fitting models of the form

$$y_i \quad = \quad f\left(x_i\right) + \epsilon_i \qquad\qquad i = 1, \ldots, n$$

Ordinary Least Squares is *optimal* under the condition

$$\epsilon_1, \ldots, \epsilon_n \text{ are i.i.d. } N\left(0, \sigma^2\right)$$

and is a reasonable fitting method when this condition is at least approximately satisfied.

Most importantly we require here that there should be **no significant outliers**.

Weighted Least Squares

If instead we have heteroscedastic data, with

$$\epsilon_1, \ldots, \epsilon_n \text{ are independent } N\left(0, \sigma_i^2\right)$$

it is natural to use instead Weighted Least Squares (WLS).

Choose $\hat{f}$ from within the permitted class of functions f as follows:

$$\text{OLS: minimise} \quad \sum_i \left(y_i - \hat{f}\left(x_i\right)\right)^2$$

$$\text{WLS: minimise} \quad \sum_i w_i \left(y_i - \hat{f}\left(x_i\right)\right)^2$$

where for simple linear regression, the class of permitted functions is $f(x_i) = a + bx_i$

The weight w_i expresses the importance we attach to observation i in fitting the model.

Natural that points with lower uncertainty are given greater weight.

In fact, it's optimal (smallest standard errors) to take w_i proportional to $1/\sigma_i^2$. But we don't usually know the true σ_i^2 values, so need to estimate in some way.

Example 7.1

Return to hill races data from chapter 6.

Seems reasonable to assume greater variability in the times for longer races — with the variability perhaps proportional to the distance.

Refit hills.model.2 with weights proportional to $1/\text{distance}^2$

$$\texttt{hills.model.3} = \texttt{lm}(\texttt{time} \sim -1 + \texttt{dist} + \texttt{I}(\texttt{dist\^{}2}) + \texttt{climb} + \texttt{I}(\texttt{climb\^{}2}), \texttt{data} = \texttt{hills}[-18,], \texttt{weights} = 1/\texttt{dist\^{}2})$$

```
> summary(hills.model.3)
Call:
lm(formula = time ~ -1 + dist + I(dist^2) + climb + I(climb^2),
    data = hills[-18, ], weights = 1/dist^2)

Weighted Residuals:
     Min       1Q   Median       3Q      Max
-2.39289 -0.39879  0.09138  0.43629  1.79996

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
dist       4.943e+00  5.567e-01   8.879 6.76e-10 ***
I(dist^2)  5.485e-02  3.817e-02   1.437  0.16114
climb      3.494e-03  2.082e-03   1.678  0.10367
I(climb^2) 1.338e-06  4.205e-07   3.182  0.00339 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.9236 on 30 degrees of freedom
Multiple R-squared:  0.9855,    Adjusted R-squared:  0.9836
F-statistic: 509.4 on 4 and 30 DF,  p-value: < 2.2e-16
```

The fitted model now is:

$$\widehat{\text{time}} = 4.94 \times \text{dist} + 0.0549 \times \text{dist}^2 + 0.00349 \times \text{climb} + 0.00000134 \times \text{climb}^2$$

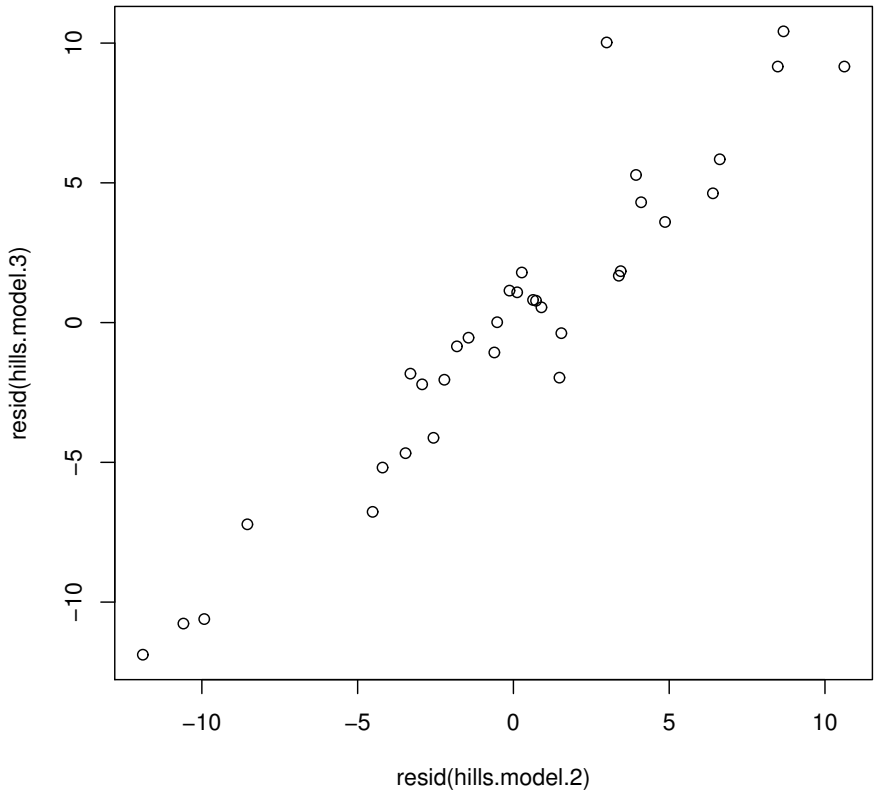
[Compare with Ordinary Least Squares fit from chapter 6:

$$\widehat{\text{time}} = 5.62 \times \text{dist} + 0.0323 \times \text{dist}^2 + 0.000262 \times \text{climb} + 0.00000180 \times \text{climb}^2$$

]

Note that although coefficients have changed from model 2 (unweighted), the fitted model is not very different.

```
plot(resid(hills.model.3) ~ resid(hills.model.2))
```



Plot confirms that the two fitted models are not very different.

Resistant regression

As already observed, Least Squares fitting is very sensitive to outlying observations.

However, there are also a large number of resistant fitting techniques available.

One such is Least Trimmed Squares (LTS): choose $\hat{f}$ from within the permitted class of functions f to minimise

$$\sum_{i=1}^{q} \left(y_i - \hat{f}\left(x_i\right) \right)^2$$

where the sum is taken only over the q smallest squared residuals, with (usually) $q = \lfloor (n + p - 1)/2 \rfloor$

This requires iterative refitting of the model until satisfactory convergence is achieved.

Example 7.2: Phones data

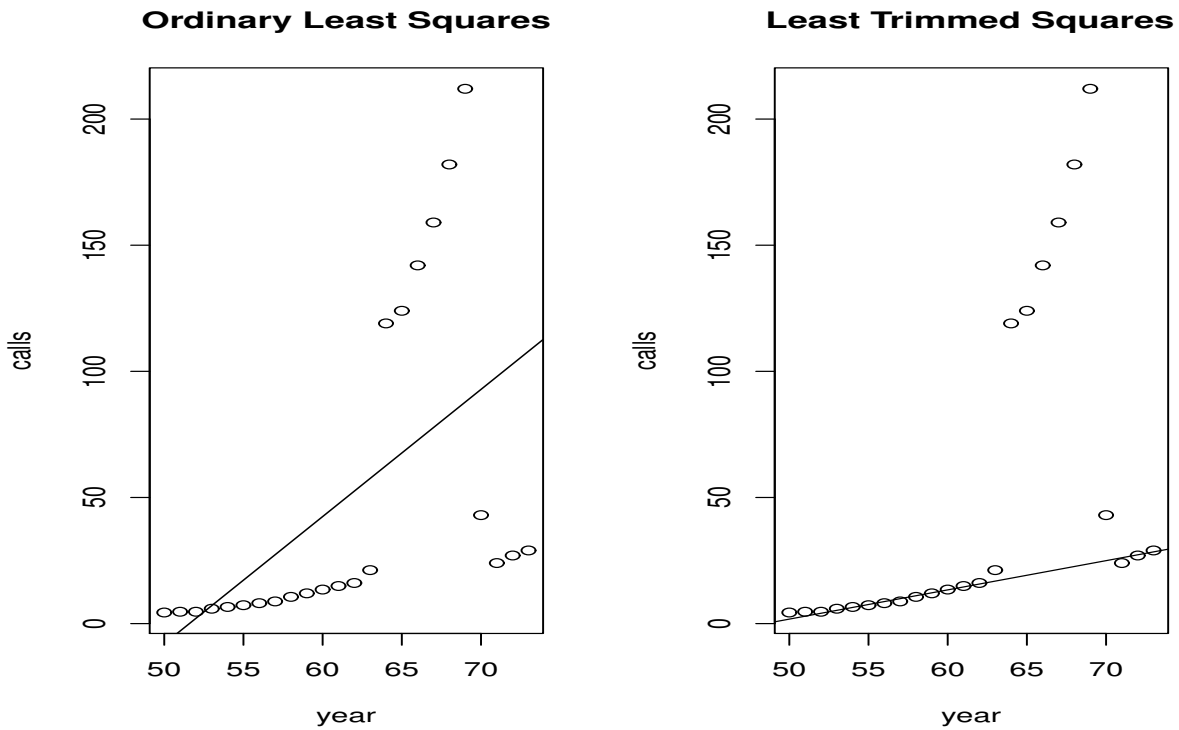
The R dataset `phones` in the package MASS gives the annual number of phone calls (millions) in Belgium over the period 1950–73.

Consider the model

$$\text{calls} \quad = \quad a + b \times \text{year}$$

The following two graphs plot the data and show the result of fitting the model by Ordinary Least Squares and then fitting the same model by Least Trimmed Squares.

```
> par(mfrow=c(1,2))
> plot(calls~year,main="Ordinary Least Squares")
> phones.ls=lm(calls~year)
> abline(phones.ls)
> plot(calls~year,main="Least Trimmed Squares")
> phones.lts=lqs(calls~year)
> abline(phones.lts)
```



There's clearly something strange about the data between 1964 and 1970.

The explanation is that for a period of time total **length** of all phone calls was accidentally recorded, instead of **number** of calls.

The incorrect data drastically affect the OLS fit, but not the LTS fit, which follows the correct data.

Nonparametric regression

Sometimes we simply wish to fit a smooth model without specifying any particular functional form for f. Again there are very many techniques here.

One such is called `loess` (LOcal regrESSion)

This constructs the fitted value $\hat{f}(x_i)$ for each observation i by performing a local regression using only those observations with x values in the neighbourhood of x_i (and attaching most weight to the closest observations).

Example 7.3: Cars data

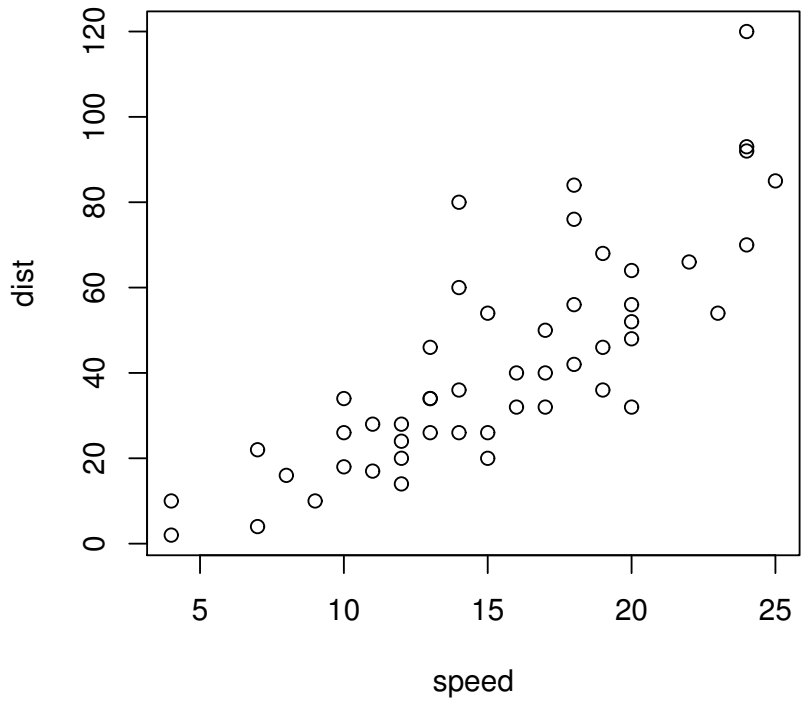
The R data frame `cars` (in the base package) records 50 observations of *speed* (mph) and *stopping distance* (ft). These observations were collected in the 1920s.

We treat stopping distance as the response variable and seek to model its dependence on speed.

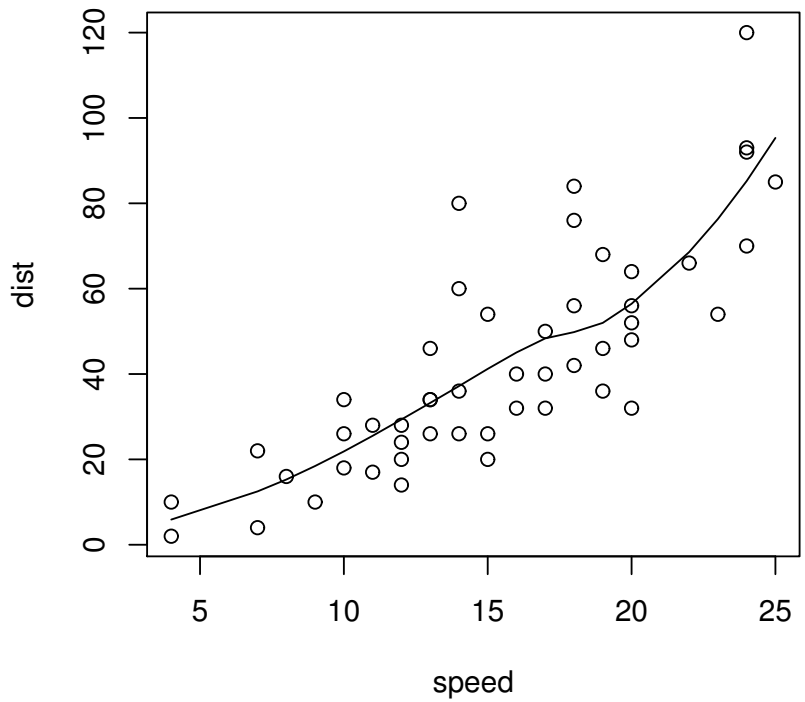
```
> detach(hills)
> data(cars)
> attach(cars)
```

(Note: `detach` the `hills` data set because both `hills` and `cars` contain a variable called `dist`.)

```
> plot(dist ~ speed)
```

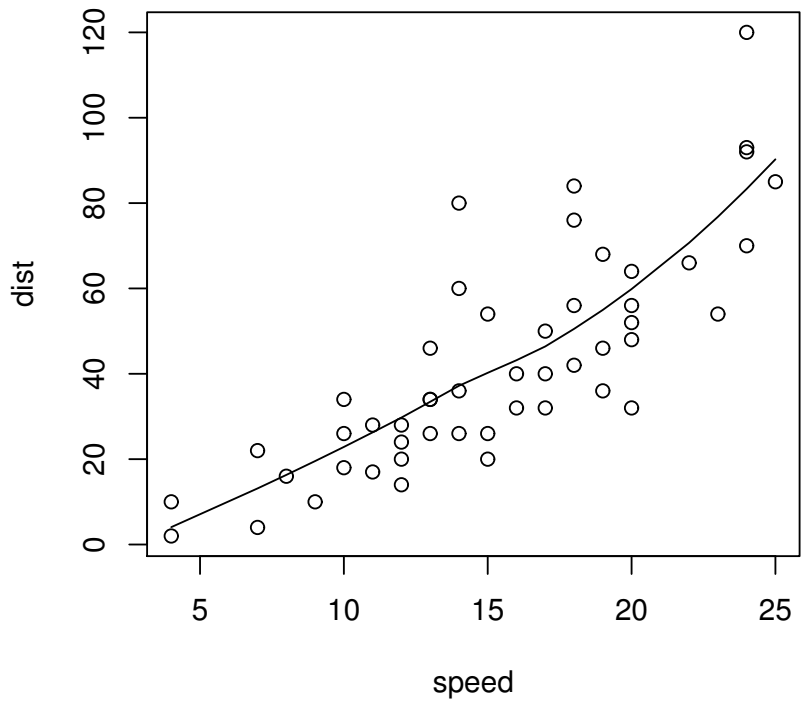


```
> cars.lo = loess(dist ~ speed)
> plot(cars)
> lines(fitted(cars.lo) ~ speed)
```



The optional argument `span` can be increased from its default value of 0.75 to give more smoothing (use `?loess` for info).

```
> cars.lo2 = loess(dist ~ speed, span=1)
> plot(cars)
> lines(fitted(cars.lo2) ~ speed)
```



More robust and resistant fits can be given by specifying the further optional argument `family="symmetric"`

Again, use `?loess` for more details.